House Prices - Ames, Iowa

Exploratory Data Analysis (EDA) Report

Data Tools and Applications - Spring 2026

HBKU

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This report presents an Exploratory Data Analysis (EDA) of the Ames, Iowa housing dataset from the Kaggle 'House Prices: Advanced Regression Techniques' competition. The dataset contains 1,460 residential property sales with 81 features describing almost every aspect of the homes. The analysis covers outlier detection, redundant attribute identification, same-value dominated attributes, and categorical merging recommendations.

Dataset: https://www.kaggle.com/c/house-prices-advanced-regression-techniques

1. Dataset Overview

The training dataset contains 1,460 rows and 81 columns.

- Numeric attributes: 38
- Categorical attributes: 43
- Target variable: SalePrice

Top Missing-Value Columns

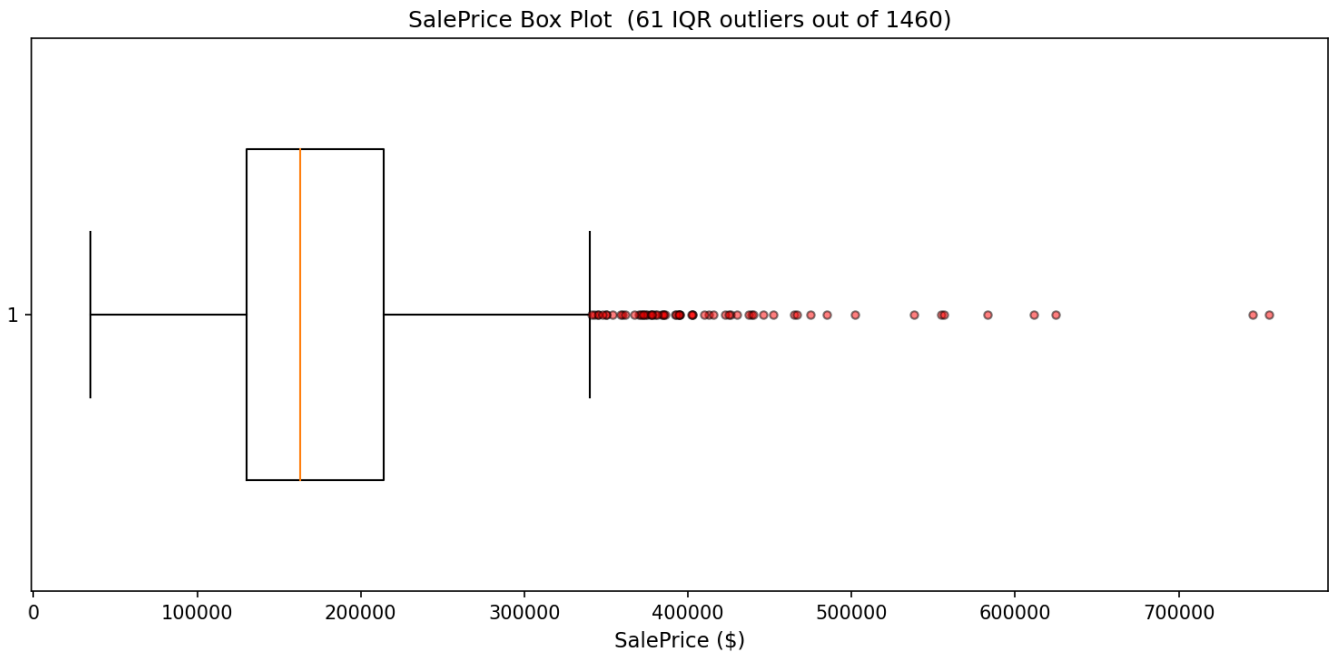
Column	Missing Count	% Missing
PoolQC	1453	99.5%
MiscFeature	1406	96.3%
Alley	1369	93.8%
Fence	1179	80.8%
FireplaceQu	690	47.3%
LotFrontage	259	17.7%
GarageType	81	5.5%
GarageYrBlt	81	5.5%
GarageFinish	81	5.5%
GarageQual	81	5.5%
GarageCond	81	5.5%
BsmtExposure	38	2.6%
BsmtFinType2	38	2.6%
BsmtFinType1	37	2.5%
BsmtCond	37	2.5%

Columns like PoolQC, MiscFeature, Alley, Fence, and FireplaceQu have very high missing rates (> 45%). Many of these represent features that simply do not exist for the property (e.g., no pool, no alley access) and are coded as NaN.

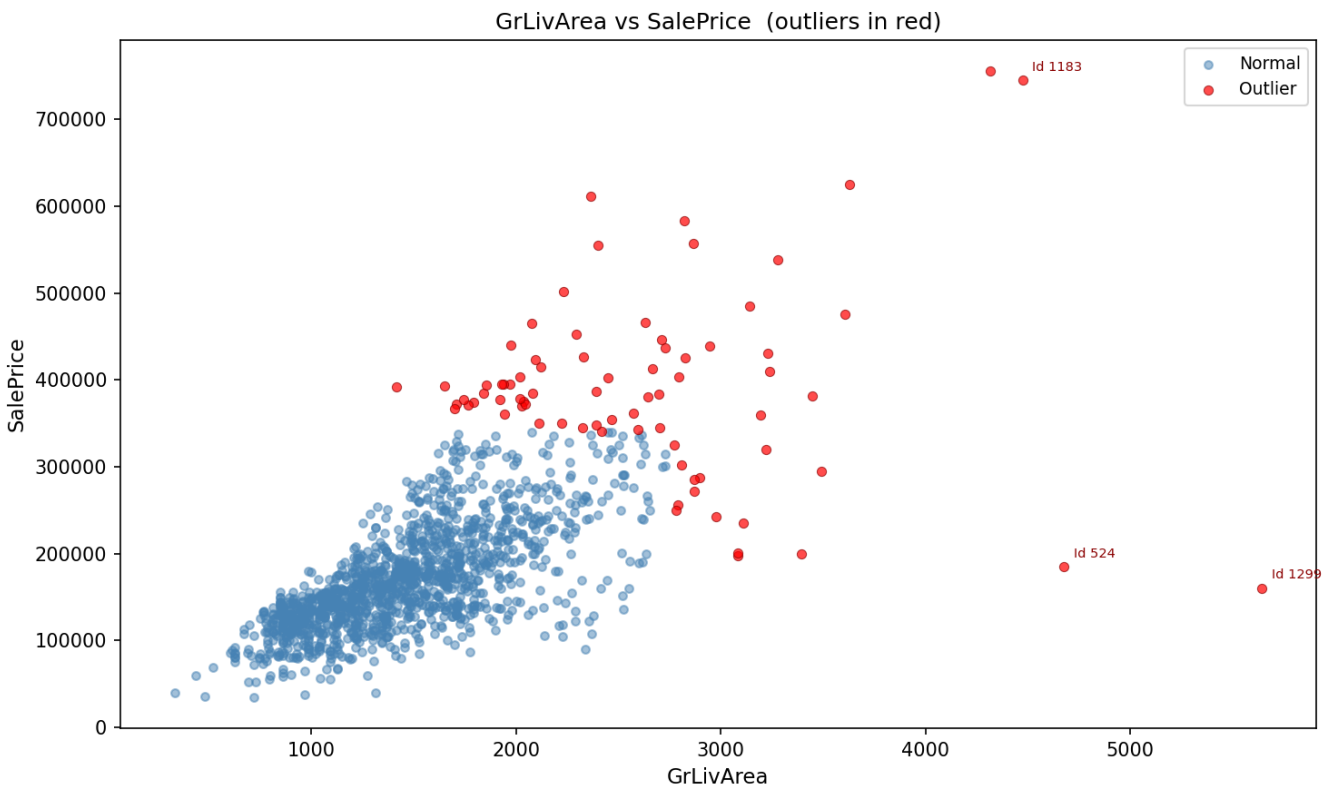
2. Outlier Detection (Task 1)

Outliers are identified using the Interquartile Range (IQR) method. A data point is considered an outlier if it falls below Q1 - 1.5*IQR or above Q3 + 1.5*IQR. Three key attributes are examined: SalePrice, GrLivArea vs SalePrice, and LotArea vs SalePrice.

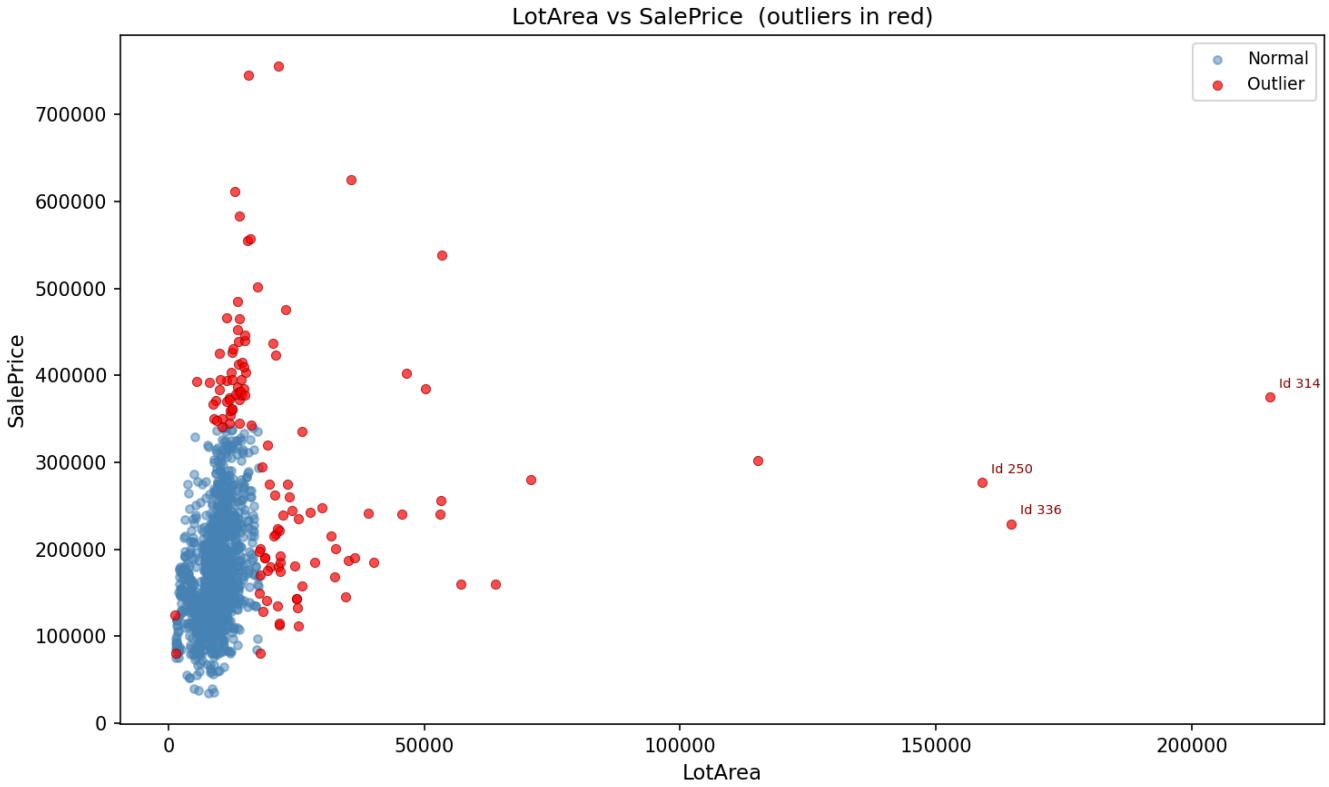
2.1 SalePrice Distribution (Box Plot)



2.2 GrLivArea vs SalePrice



2.3 LotArea vs SalePrice



2.4 Outlier Summary

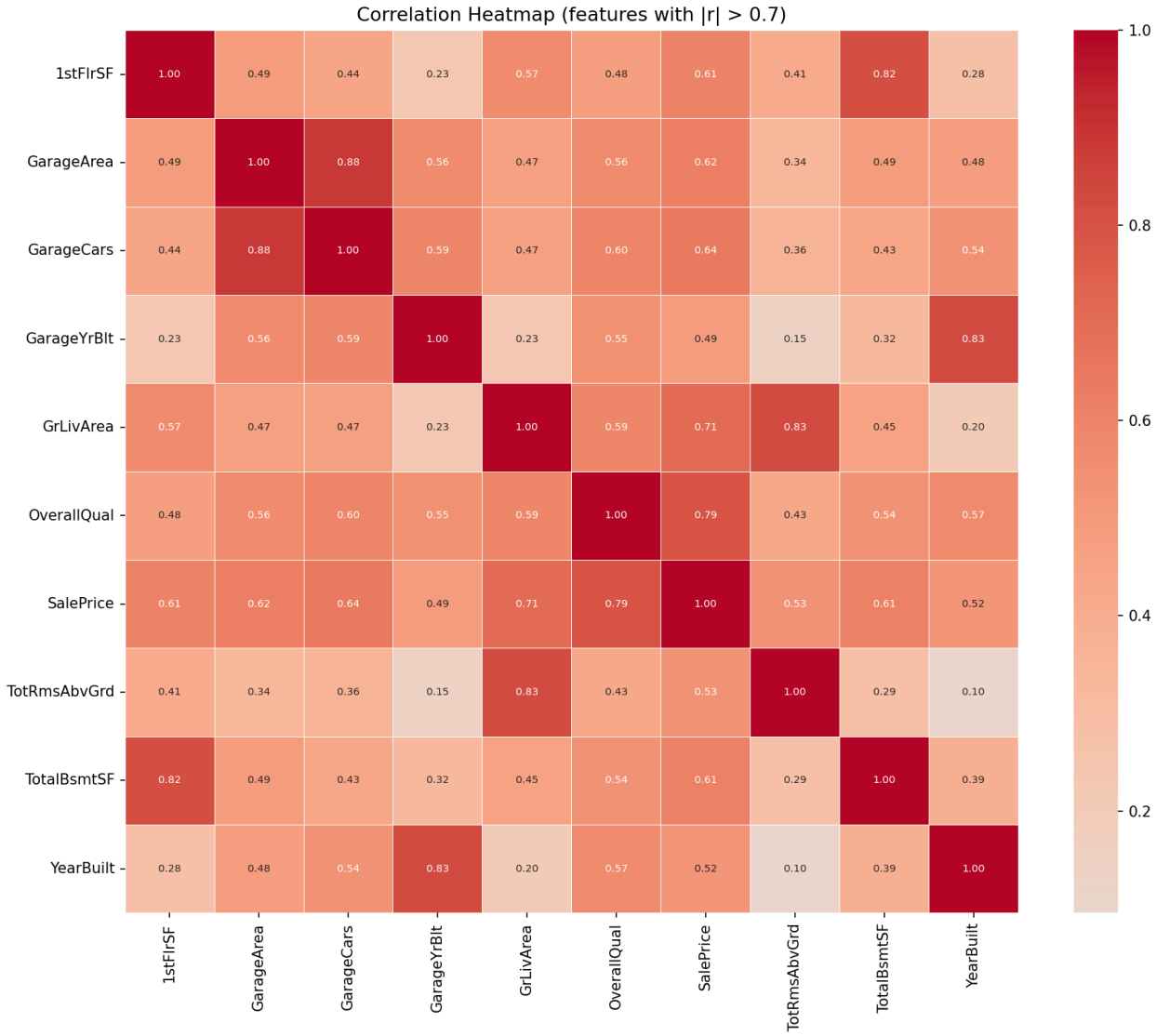
Attribute	Outlier Count	% of Data	Lower Bound	Upper Bound
SalePrice	61	4.2%	3,938	340,038
GrLivArea	31	2.1%	159	2,748
LotArea	69	4.7%	1,482	17,674

SalePrice has the most outliers by count, concentrated on the upper end (expensive homes). LotArea has a handful of extremely large lots that could heavily influence regression models. The two large GrLivArea outliers (~4,500+ sq ft) with low sale prices are well-known anomalies often removed in modeling.

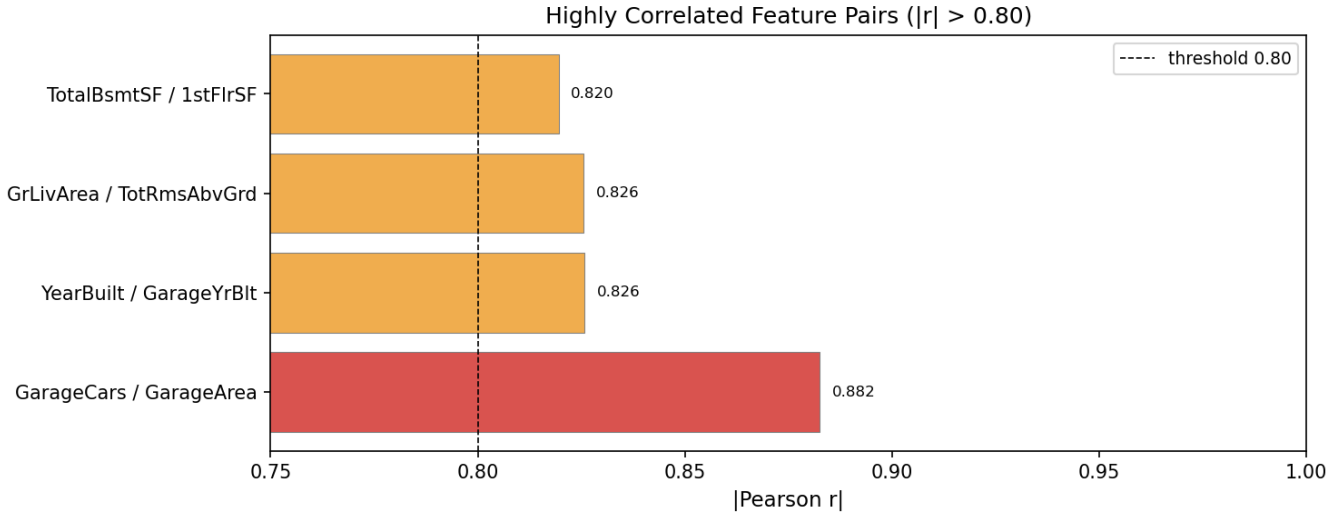
3. Redundant Attributes (Task 2)

Redundant attributes are identified using Pearson correlation. Pairs of numeric features with |r| > 0.80 are flagged as potentially redundant, meaning one could be dropped without significant information loss.

3.1 Correlation Heatmap (features with |r| > 0.7)



3.2 Highly Correlated Feature Pairs



3.3 Redundant Pairs (|r| > 0.80)

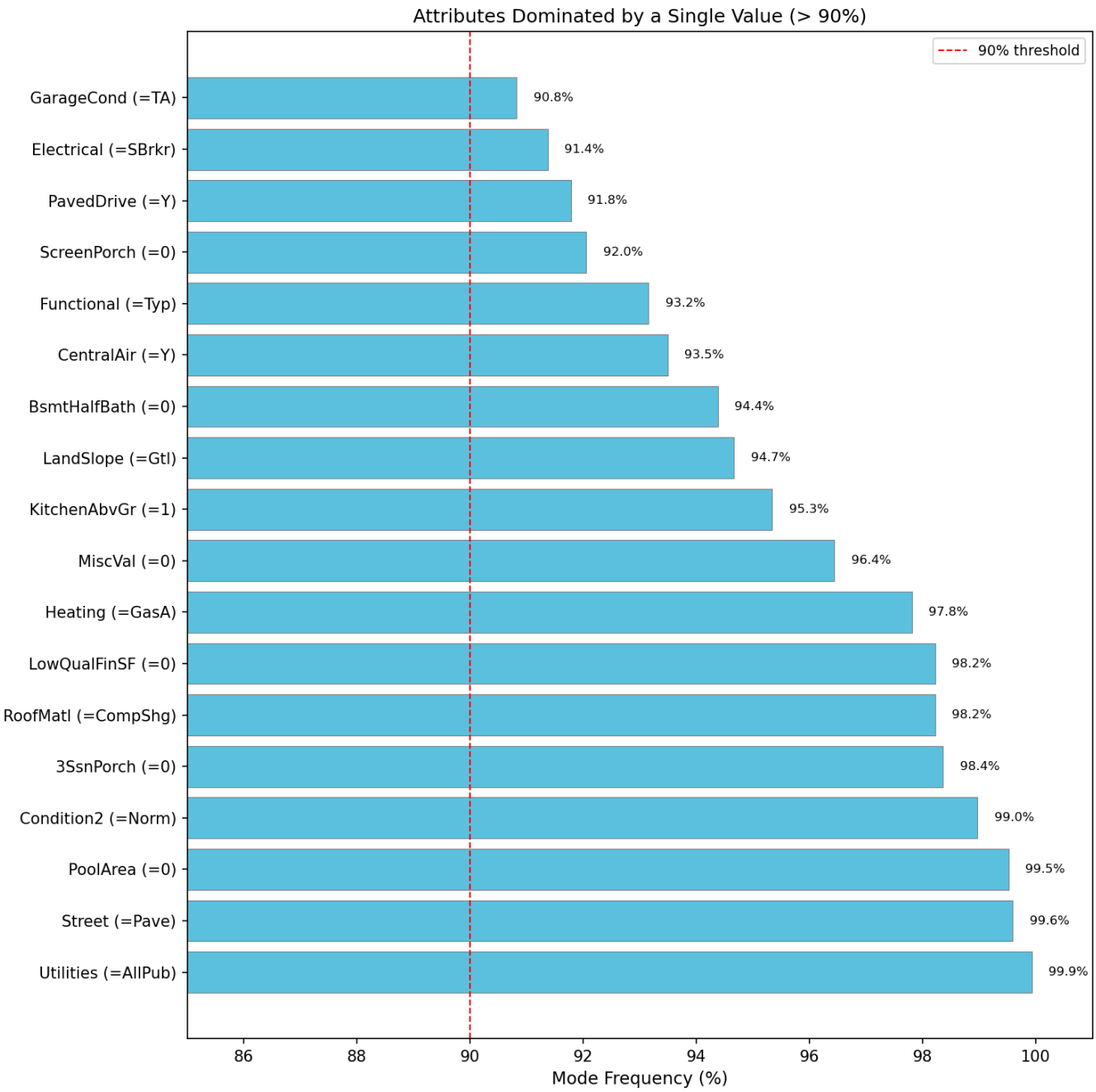
Feature A	Feature B	Pearson r	Recommendation
GarageCars	GarageArea	0.8825	Keep GarageCars (more interpretable, fewer outliers).
YearBuilt	GarageYrBlt	0.8257	Keep YearBuilt (fewer missing values).
GrLivArea	TotRmsAbvGrd	0.8255	Keep GrLivArea (continuous, more precise).
TotalBsmtSF	1stFlrSF	0.8195	Keep TotalBsmtSF (captures full basement info).

The strongest redundancy is between GarageArea and GarageCars (r ~ 0.88). Keeping one from each pair reduces multicollinearity without losing predictive power.

4. Same-Value Dominated Attributes (Task 3)

Attributes where a single value accounts for more than 90% of all observations provide very little discriminative power and are candidates for removal.

4.1 Attributes with > 90% Same Value



4.2 Flagged Attributes

Attribute	Dominant Value	Frequency %
Utilities	AllPub	99.9%
Street	Pave	99.6%
PoolArea	0	99.5%

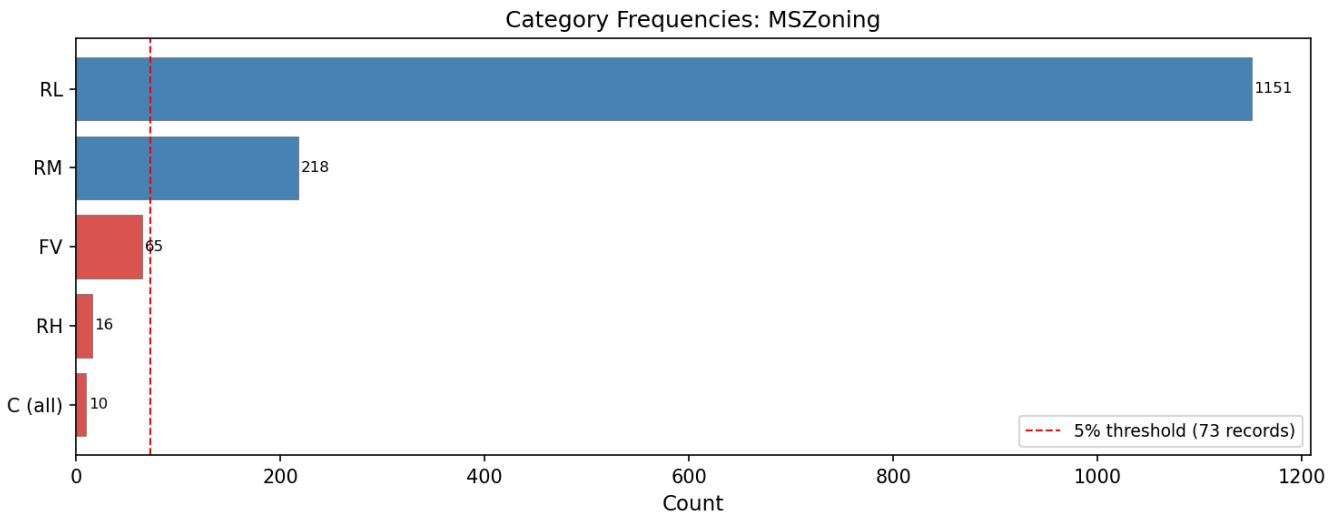
Condition2	Norm	99.0%
3SsnPorch	0	98.4%
RoofMatl	CompShg	98.2%
LowQualFinSF	0	98.2%
Heating	GasA	97.8%
MiscVal	0	96.4%
KitchenAbvGr	1	95.3%
LandSlope	Gtl	94.7%
BsmtHalfBath	0	94.4%
CentralAir	Y	93.5%
Functional	Typ	93.2%
ScreenPorch	0	92.0%
PavedDrive	Y	91.8%
Electrical	SBrkr	91.4%
GarageCond	TA	90.8%

Utilities is almost entirely 'AllPub' (99.9%) and carries virtually zero information. Street (99.6% 'Pave'), PoolArea (99.5% value 0), and several condition/functional columns are similarly dominated. These attributes can typically be dropped from predictive models without impact.

5. Categorical Merging Analysis (Task 4)

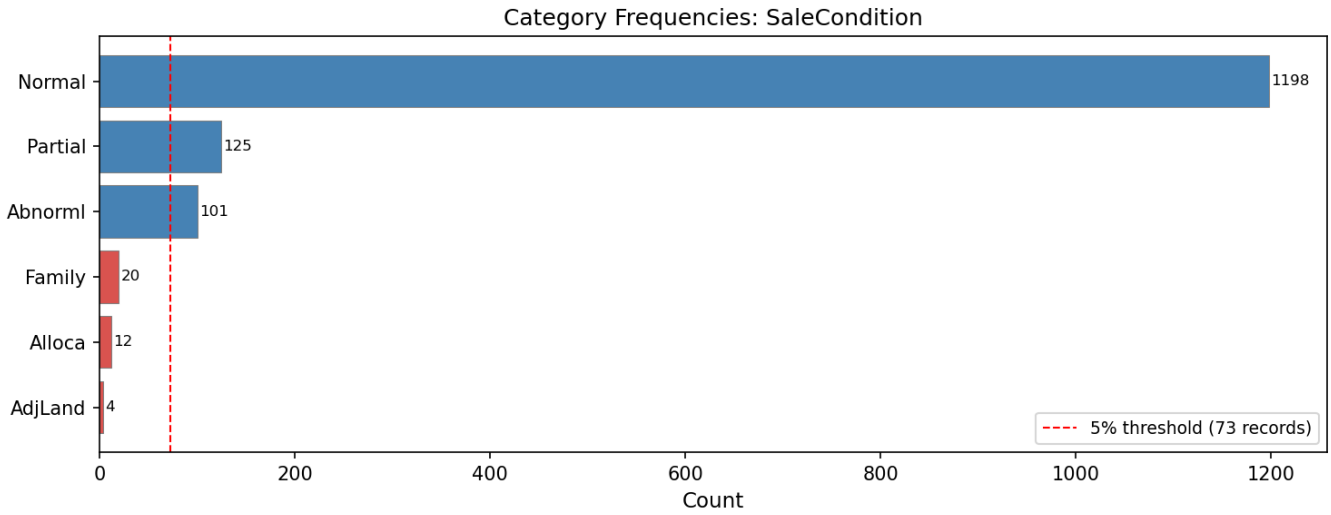
For categorical attributes, categories with fewer than 5% of observations (< 73 records) are flagged as merge candidates. Merging rare categories into an 'Other' group reduces sparsity and improves model stability.

5.1 MSZoning - Category Frequencies



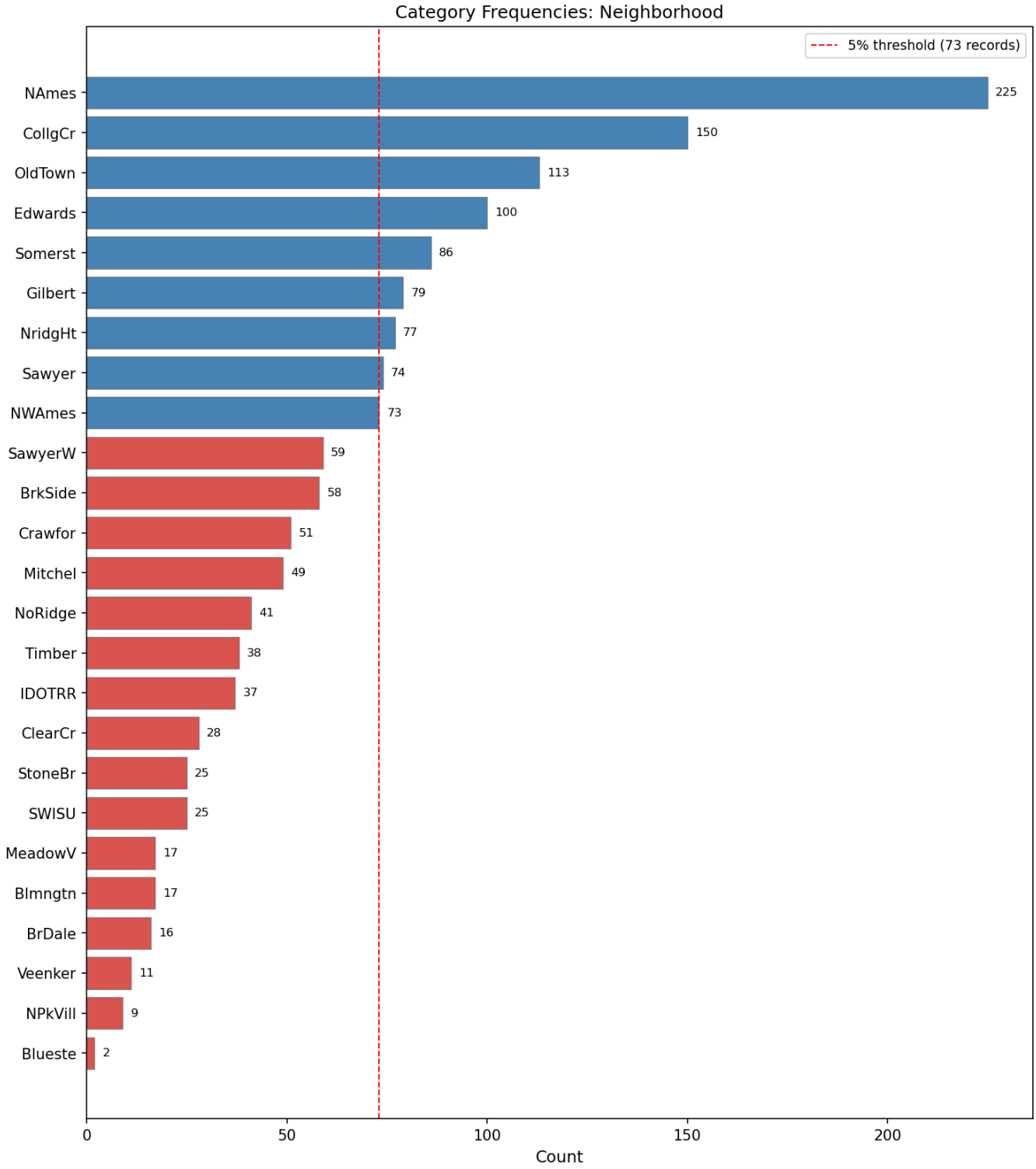
Merge candidates (< 5%): FV (4.5%), RH (1.1%), C (all) (0.7%). Recommendation: merge these into 'Other' or combine with the closest zoning type (e.g., RH and RM could merge into 'Residential_Medium').

5.2 SaleCondition - Category Frequencies



Merge candidates (< 5%): Family (1.4%), Alloca (0.8%), AdjLand (0.3%). Recommendation: merge AdjLand, Alloca, and Family into 'Other_Condition'.

5.3 Neighborhood - Category Frequencies



Merge candidates (< 5%): 16 neighborhoods fall below the threshold. Recommendation: group rare neighborhoods by geographic proximity or median SalePrice similarity into broader clusters (e.g., 'Low_Density', 'Mid_Range').

6. Conclusions

1. Outlier Detection: The IQR method identified outliers in SalePrice, GrLivArea, and LotArea. Two well-known GrLivArea outliers (large area, low price) should be removed before modeling. Extreme LotArea values may also warrant capping or removal.

2. Redundant Attributes: Four pairs of features have |r| > 0.80: GarageArea/GarageCars, TotalBsmtSF/1stFlrSF, GrLivArea/TotRmsAbvGrd, and YearBuilt/GarageYrBlt. Dropping one feature from each pair reduces multicollinearity with minimal information loss.

3. Same-Value Attributes: Several columns are dominated by a single value (> 90%), most notably Utilities (99.9% AllPub), Street (99.6% Pave), and PoolArea (99.5% zero). These provide negligible predictive value and can be safely removed.

4. Categorical Merging: Rare categories exist in MSZoning, SaleCondition, and especially Neighborhood. Merging these below-threshold categories reduces sparsity and prevents overfitting during one-hot encoding.

These preprocessing steps form a solid foundation for subsequent feature engineering and predictive modeling on the Ames housing dataset.